

Networks and Systems: MM11

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- **Mm1-7** Network Performance Analysis (L2-L5) + CAN/Flexray + clocks
- **Mm8-9** Discrete Event Simulation (HS)
- **Mm10** Dependability Modeling via Markov Chains (hps)
- **Mm11** Fault detection via Hidden Markov Chains (hps)

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Purpose of lecture

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- Understand basic properties of fault-detectors
 - ▣ Metrics: false/true positives/negatives, reactivity
 - ▣ Understand Trade-off between metrics
 - How to create and read ,ROC' curves
- Be able to design and apply HMM-based fault-detectors
 - ▣ Know the **elements** that comprise an HMM
 - ▣ Be able to **solve** simple discrete time HMM problems
 - Become familiar with **Forward** and **Viterbi** algorithms and how they work
 - ▣ Know how to map a fault-detection problem to an HMM-based algorithm

Fault-detection - definition

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- Fault-detector is a function, $fd(t)$
 - ▣ Taking system observations $o(t)$ as input (frequently sampled in discrete time, $o(t_i)$);
 - $o(t)$ can also be a sequence of observations collected until time t
 - ▣ Output, $fd(t)$, representing the conjecture about the presence of a fault
 - Binary: 1=fault present, 0 otherwise
 - $fd(t)$ in $[0,1]$: value representing estimated probability of presence of fault
 - ▣ Extensions with/to diagnosis component: type of fault, transient/persistent, etc.
- Ideal fault-detector:
 - ▣ Output=1, if and only if the fault is present, otherwise 0

Overview

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- Fault-detectors: examples and quality metrics

True/false positives, True/false negatives, reactivity

Trade-off in parameterization, RoC curves

- Introduction of Hidden Markov Models

- ▣ Motivation for HMMs

- ▣ Elements of a HMM

- Efficient algorithms for HMMs

- ▣ Three basic HMM problems

- ▣ Algorithms

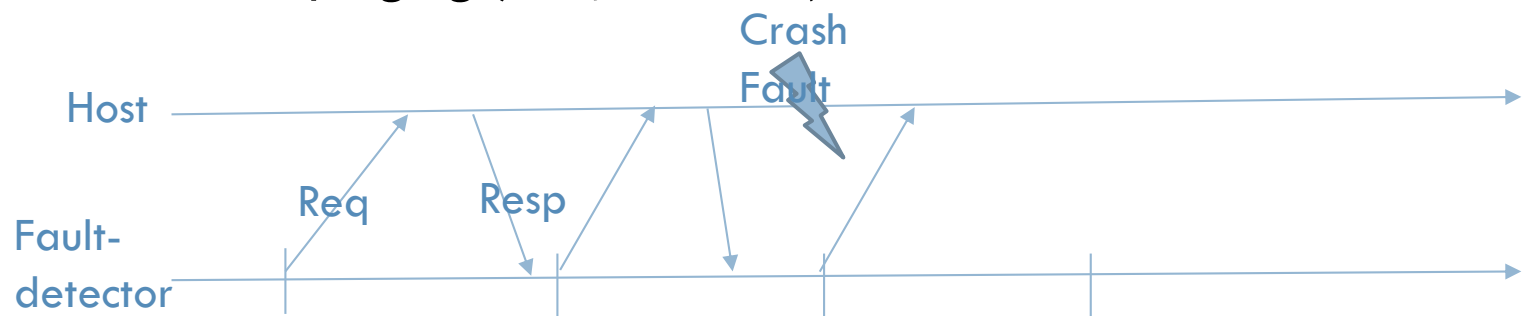
- ▣ Use of HMMs for fault-detection

- Exercises

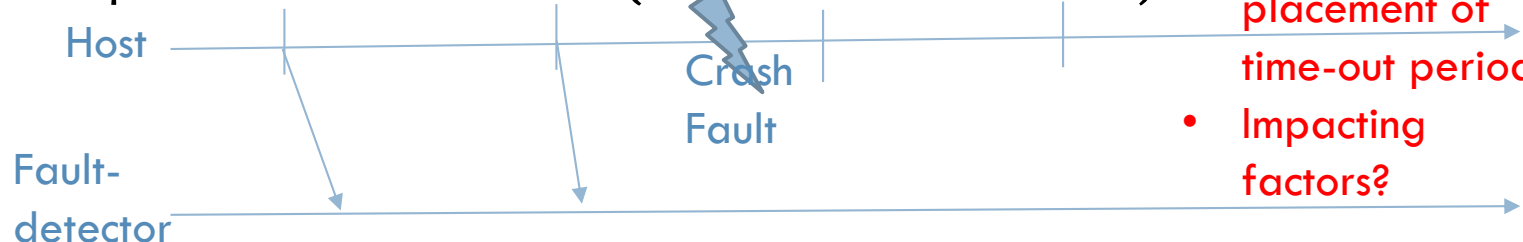
Example: Detection of node crashes

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- Purpose: Check if another host is up/down.
- Methods – time-out based
 - ▣ Pull: Periodic pinging (are you alive?)



- ▣ Push: proactive notifications (I am alive - heartbeats)



Discussions:

- Exact placement of time-out period!
- Impacting factors?

True and False Positives and Negatives

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4 Cases (shown as 'Confusion Matrix')

(2 system states, S , and 2 fault detector outputs, FD)

Negatives

$FD=0$

Positives

$FD = 1$

$S=0$: $TNR := \Pr(FD=0 \mid S=0)$ $FPR := \Pr(FD=1 \mid S=0)$

$S=1$: $FNR := \Pr(FD=0 \mid S=1)$ $TPR := \Pr(FD=1 \mid S=1)$

Consequence: $TN=1-FP$; $FN=1-TP$;

Legend: TNR = True Negative Rate, FPR = False Positive Rate, etc.

Accuracy

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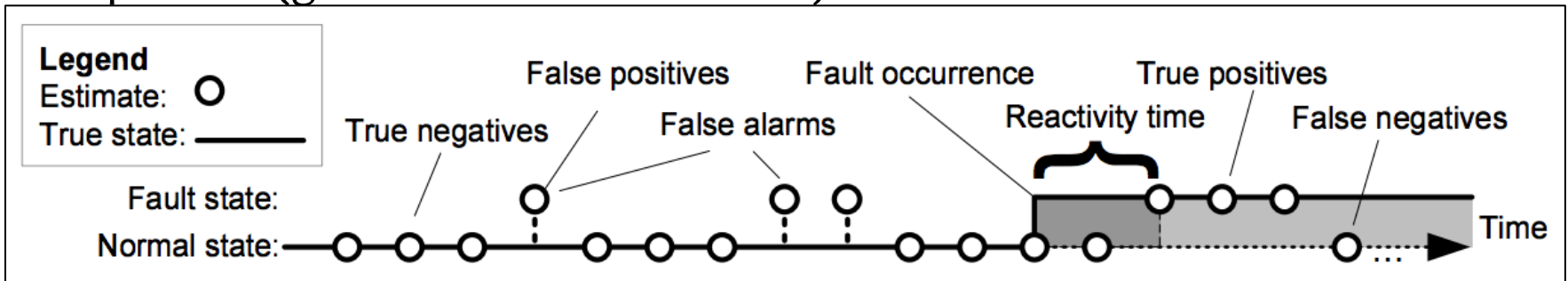
$$\begin{aligned}\text{Accuracy} &:= \Pr(\text{Fdetector} = \text{SystemState}) \\ &= \text{TPR} * \Pr(S=1) + \text{TNR} * \Pr(S=0)\end{aligned}$$

Note: In the general case, accuracy depends on how often (and how long) the system is faulty

Fault-detector - metrics

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- Accuracy: probability that fault-detector shows correct result
 - ▣ At moments of estimation versus at any moment in time?
 - ▣ More detailed metrics: Probability of false positives and false negatives
- Reactivity: Distribution of time between fault occurrence and first true positive (given the fault is detected)



From: Nickelsen, A.; Gronbaek, J.; Renier, T.; Schwefel, H.-P.; , "Probabilistic Network Fault-Diagnosis Using Cross-Layer Observations," Advanced Information Networking and Applications, 2009. AINA '09. International Conference on.

→ Calculate these metrics for the heart-beat detector from the previous slide

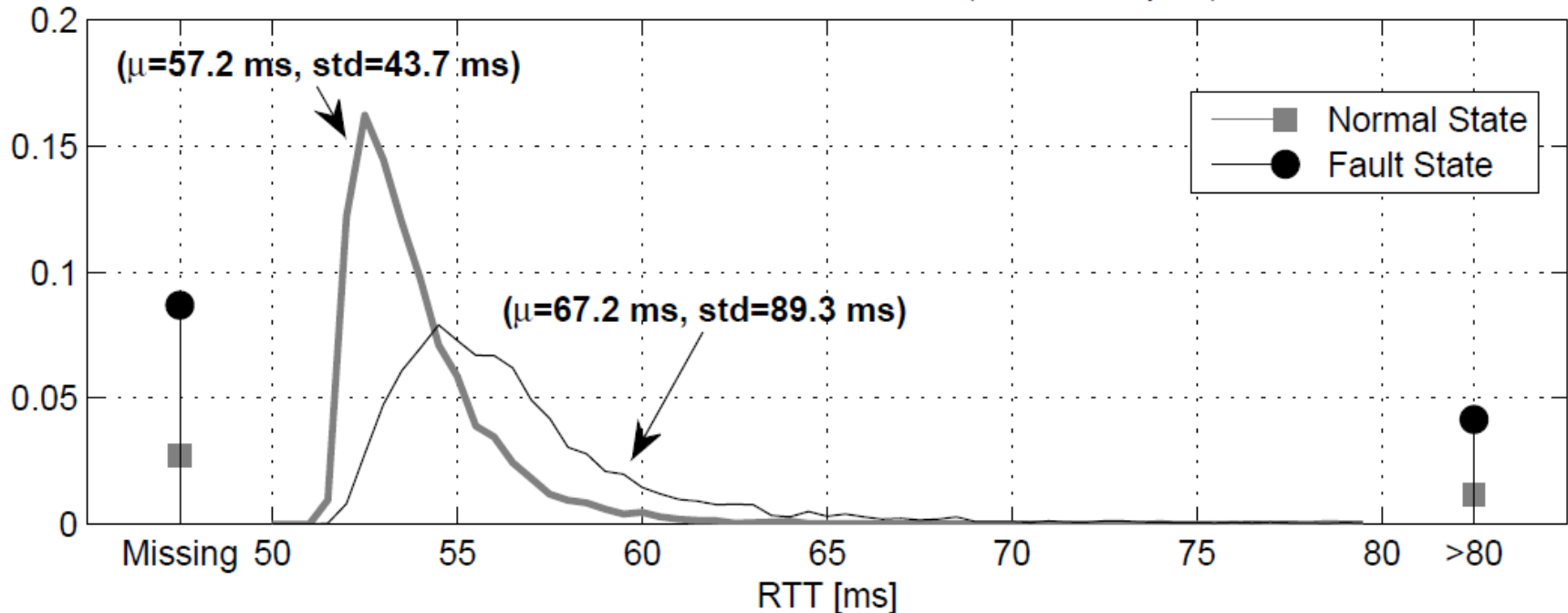
Example: congestion detection

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- Another example: congestion detection based on round-trip time delays (RTT)

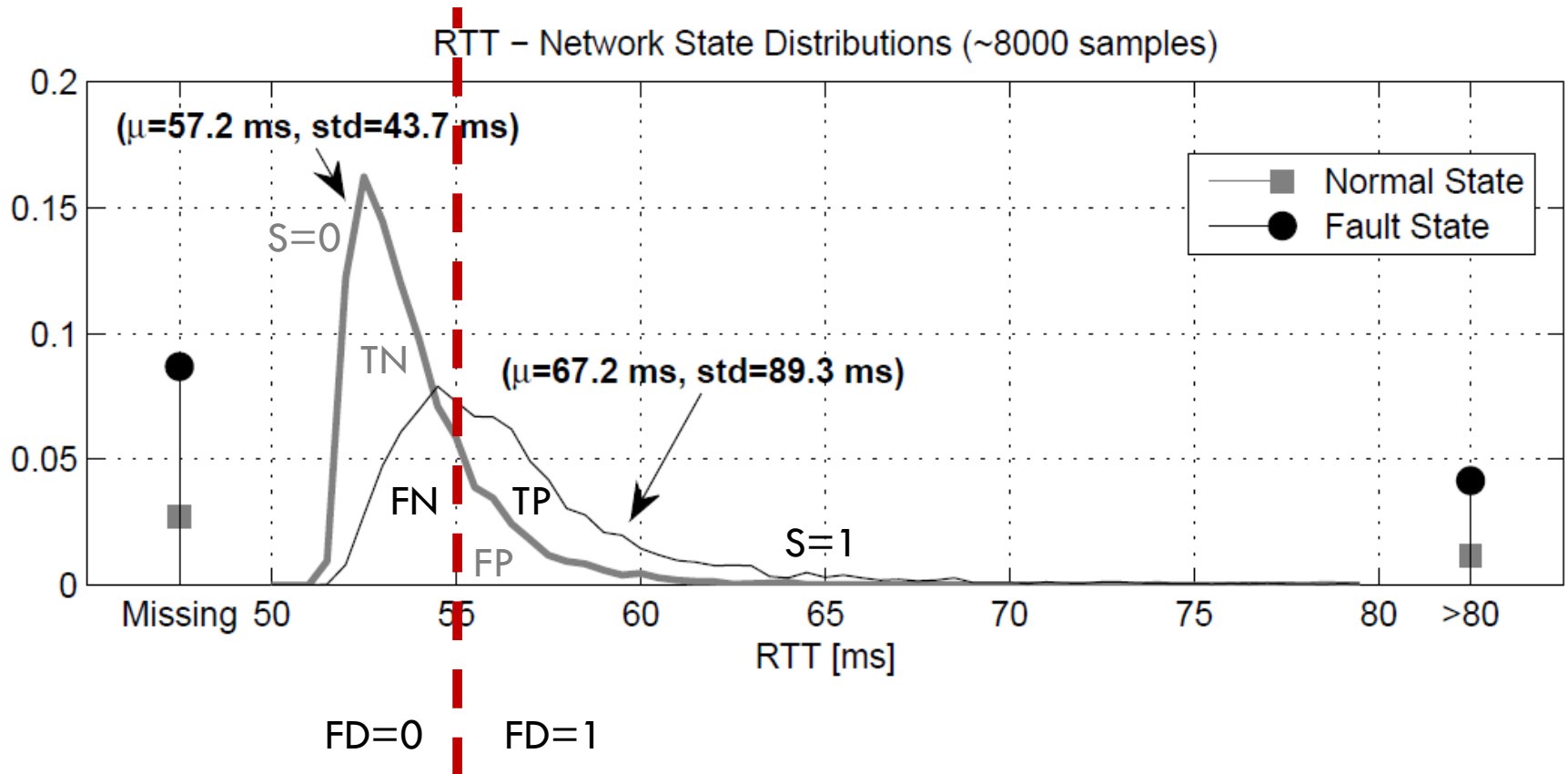
- $FD(t_i) = 1(RTT_i > \text{threshold})$

RTT - Network State Distributions (~8000 samples)



Deriving the confusion matrix – simple threshold detector

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Example: Thresholds = 55ms

ROC curve (Receiver Operating Characteristic)

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- E.g. for threshold-based congestion detector, increasing threshold

- ▣ Decreases rate of false positives:

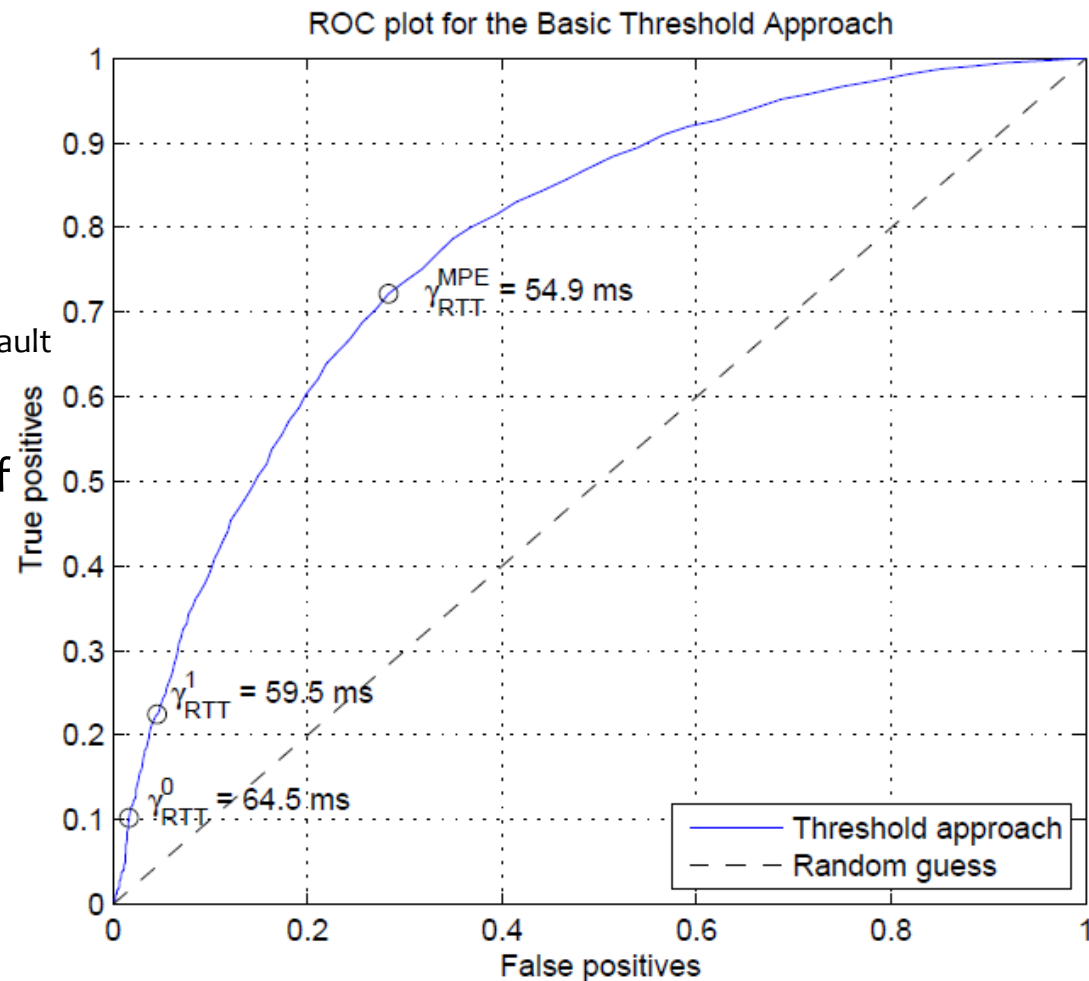
- # false positives / all cases of no fault
- $\Pr(\text{FP} \mid \text{normal operation})$

- ▣ But also decreases rate of true positives

- # true positives / all cases of faults
- $\Pr(\text{TP} \mid \text{faulty operation})$

Plotting the resulting curve is called a ROC curve

- Represents behavior of detector for varying parameterization



Discuss 'meaning' of corners of sector and different ROC curve placements

Exercises 1/2

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1. Congestion detection: An end-to-end communication path in ‘good’ condition shows a round-trip time which is exponential with mean 500ms. During congestion cases, the distribution becomes an exponential with mean 800ms.
 - ▣ Plot the density of the two distributions
 - ▣ A threshold detector is used to detect congestion cases. Plot the following curves:
 - The true positive rate over the threshold
 - The false positive rate over the threshold
 - The ROC curve
 - The accuracy of the estimator over the threshold, assuming that the true state of the network is in 70% of the cases ‘good’; plot in the same graph the accuracy of the estimator if the network is only in 40% of the cases in the ‘good’ condition.

References

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- Basic networking: A. Tanenbaum, Computer Networks, 4rd.
- Jesper Grønbæk: Decentralized Fault Management for Service Dependability in Ubiquitous Networks, PhD Thesis, Aalborg University, October 2010.
- A. Bondavalli, S. Chiaradonna, F. Di Giandomenico, and F. Grandoni. Threshold-based mechanisms to discriminate transient from intermittent faults. IEEE Transactions on Computers, 49(3):230–245, 2000
- Anders Nickelsen, Jesper Gronbak, Thibault Renier, and Hans Peter Schwefel. Probabilistic Fault-Diagnosis in Mobile Networks Using Cross-Layer Observations. In Proceedings of AINA 2009, Bradford, UK, May 26-29, 2009.
- R.C. Nunes and I. Jansch-Porto. QoS of Timeout-Based Self-Tuned Failure Detectors: The Effects of the Communication Delay Predictor and the Safety Margin. In Proceedings of the 2004 International Conference on Dependable Systems and Networks, pages 753–761. IEEE Computer Society, 2004.

Overview

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- Fault-detectors: examples and quality metrics
 - ▣ True/false positives, True/false negatives, reactivity
 - ▣ Trade-off in parameterization, RoC curves
- Introduction of Hidden Markov Models
 - Motivation for HMMs
 - Elements of a HMM
- **Efficient algorithms for HMMs**
 - ▣ Three basic HMM problems
 - ▣ Algorithms
 - ▣ Use of HMMs for fault-detection
- Exercises

Discrete Time Markov Processes (Revision)

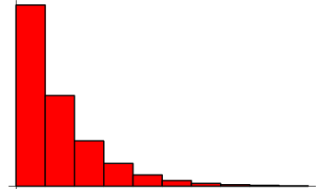
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□ Definition

- Discrete time: a transition takes constant time; not exponential as continuous time
- State-space: finite or countable infinite
E.g., $E=\{1,2,\dots,N\}$ ($N=\infty$ also allowed)
- Transition probabilities: $p_{jk} = \Pr(\text{transition from state } j \text{ to state } k)$
 - Transition probability matrix $\underline{P}$ contains all p_{jk}
- X_i = RV indicating the state of the Markov process in step i
- Markov property: State in step i only depends on state in step $i-1$:
 - $\Pr(X_i=s \mid X_{i-1}=s_{i-1}, X_{i-2}=s_{i-2}, \dots, X_0=s_0) = \Pr(X_i=s \mid X_{i-1}=s_{i-1})$

□ Computation of state probabilities

- Initial state probabilities (step 0): $\underline{\pi}_0$
- Probability of state sequence $s_0, s_1, \dots, s_i$: $\Pr(X_0=s_0, X_1=s_1, \dots, X_i=s_i) = \dots$ (see next)
- $\Pr(X_i=k) = \sum_j [\Pr(X_{i-1}=j) \cdot p_{jk}]$
- $\underline{\pi}_i = \underline{\pi}_{i-1} \underline{P}$
- State holding time: geometric with parameter p_{ii}



□ Example: 2-state MC (state = link behavior at packet tx = {ideal, erroneous})

Revision: Discrete Time Markov Processes

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□ Properties

- ▣ Irreducibility: each state is reachable from any other state (potentially in multiple steps)
 - Transient state: a state that might never be reached again
- ▣ Positive recurrence: can return to any given state in a finite number of steps
- ▣ Periodic: returns to a given state in multiple of k steps

□ Steady-state probabilities

- ▣ $\underline{\pi} = \lim_{i \rightarrow \infty} \underline{\pi}_i$
- ▣ Limit exists and is independent of $\underline{\pi}_0$ if Markov chain irreducible and aperiodic
- ▣ Aperiodic & positive recurrent = ergodic $\rightarrow \underline{\pi}$ is probability distribution,
obtained as normalized left eigenvector $\underline{\pi} = \underline{\pi} P$

Wireless Link Example (1)

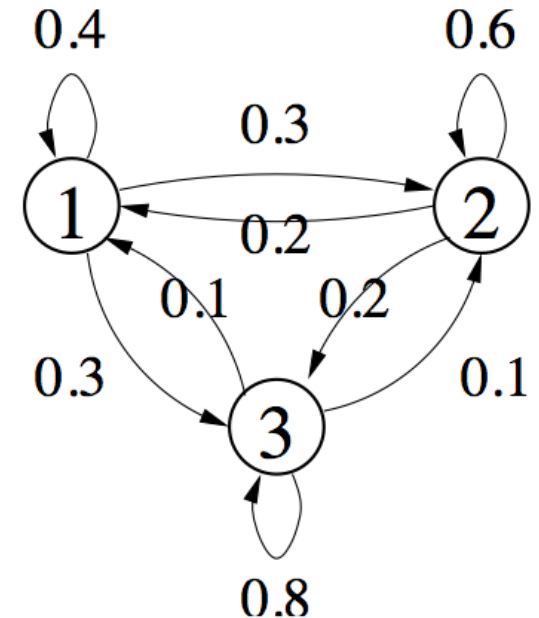
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e.g

- State 1 = good link conditions
- State 2 = intermediate link conditions
- State 3 = poor link conditions

Time step is for example 1 second.

Transition probabilities:



$$\underline{P} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

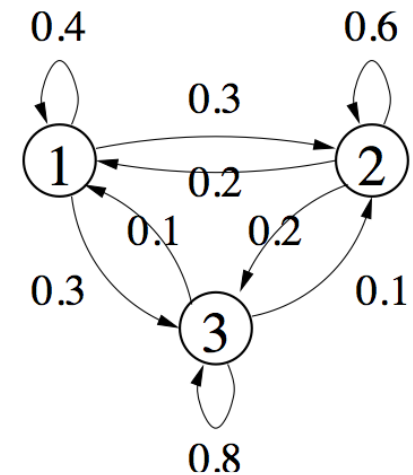
Wireless Link Example (2)

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- Given the link is in poor condition right now (i.e., $x_1 = 3$), what is the probability of “poor-poor-good-i/m-i/m-poor” with model M ?

the initial state probability for state i is assumed to be $\pi_0 = [0, 0, 1]$

$$\begin{aligned}
 P(X|\mathcal{M}) &= P(X = \{3, 3, 1, 2, 2, 3\}|\mathcal{M}) \\
 &= P(x_1 = 3) P(x_2 = 3|x_1 = 3) \\
 &\quad \pi_0^3 P(x_3 = 1|x_2 = 3) P(x_4 = 2|x_3 = 1) \\
 &\quad P(x_5 = 2|x_4 = 2) P(x_6 = 3|x_5 = 2) \\
 &= \pi_0(3) p_{33} p_{31} p_{12} p_{22} p_{23} \\
 &= 1 \cdot (0.8)(0.1)(0.3)(0.6)(0.2) \\
 &= 0.00288
 \end{aligned}$$



State 1 = good

State 2 = intermediate (i/m)

State 3 = poor

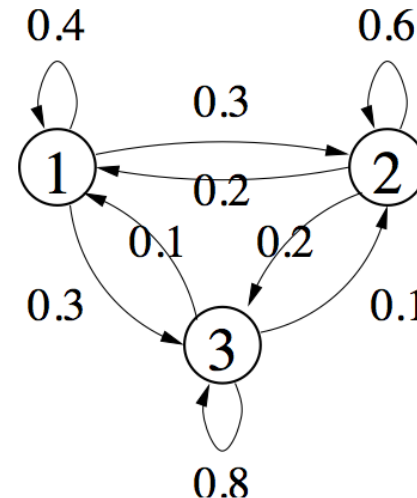
Motivation for HMMs

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Example: Wireless Link

- The different link conditions influence frame error rates on the Layer-2 transmission (Markov modulated error rates), e.g.
 - ▣ $\text{FER}(\text{good}) = 1\text{e-}5$, $\text{FER}(\text{i/m}) = 0.02$, $\text{FER}(\text{poor}) = 0.2$
- Assume that the sender sends a Layer-2 frame every second (e.g. for the example of a sensor device)
 - ▣ What can you say about the link states if the following patterns are observed
 - CCCC, EEEE, CCCE
- HMM allows us to, estimate the actual process state (here link state) from observations in a formalized way!
 - ▣ ... so can be used as fault-detectors

State 1 = good
State 2 = intermediate (i/m)
State 3 = poor



Hidden Markov Models (HMMs): Definition

- Main property
 - In each state $s \in E$, an 'observation symbol' from some alphabet V is generated probabilistically
 - The underlying state cannot be observed, only the sequence $\underline{O}=[O_1, O_2, \dots, O_T]$ of generated symbols
- $HMM = \langle E, V, \underline{\pi}_1, \underline{P}, \underline{B} \rangle$
 - E : state-space (discrete, finite/infinite), w/o.l.g. $E=\{1,2,\dots,N\}$
 - V : set of possible observation symbols (discrete for now), w/o.l.g. $V=\{1,2,\dots,M\}$
 - $\underline{\pi}_1$: initial state probabilities at step 1
 - $\underline{P}$: $N \times N$ matrix of state transition probabilities $p_{ij} = \Pr(X_{k+1}=j \mid X_k=i)$
 - $\underline{B}$: $N \times M$ matrix of symbol generation probabilities: $b_{ij} = \Pr(O_k=j \mid X_k=i)$
- **Example:** 2-state HMM, observations = result from biased coin-toss
- **Note:** Discrete time Markov model is special case of HMM, namely each column of $\underline{B}$ contains at most one non-zero element
- **Exercise:** Write a (Matlab) program with input $(\underline{\pi}_1, \underline{P}, \underline{B}, T)$ that generates a sequence of observations of length T

Matlab implementation

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```
function [S,Q]=HMM_simul(pi1,P,B,T)
current_state=select_state(pi1);
S=zeros(1,T);
Q=S;
for i=1:T,
    Q(i)=current_state;
    S(i)=select_state(B(current_state,:));
    current_state=select_state(P(current_state,:));
end;
```

```
function nr=select_state(pi)
a=rand;
for i=1:length(pi)
    a=a-pi(i);
    if a<0,
        nr=i; return;
    end;
end;
nr=-1; % pi no probability vector
```

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- **Efficient algorithms for HMMs**
 - Three basic HMM problems
 - Algorithms
 - Use of HMMs for fault-detection
- Exercises

HMM – Computations: Observation Probability

□ Problem 1: Compute probability of observing a certain sequence $\underline{o}=[o_1, \dots, o_T]$ in a given HMM.

□ First (inefficient) approach ('brute-force'):

- Generate all possible state-sequences of length T: $\underline{q}=[q_1, \dots, q_T]$
- Sum up all $\Pr(\underline{o} | \underline{q})$ weighted by $\Pr(\underline{q})$ (total probabilities)
- Problem: Number of paths grows exponentially as N^T

□ More efficient (quadratic in N) approach: forward procedure

- Iterative method computing probabilities for pre-fixes of the observation sequence:
 $\underline{\alpha}_t := [\Pr(O_1=o_1, \dots, O_t=o_t, \mathbf{X}_t=1), \dots, \Pr(O_1=o_1, \dots, O_t=o_t, \mathbf{X}_t=N)]$
- At step $t=1$: $\underline{\alpha}_1(i) = \Pr(O_1=o_1, X_1=i) = \underline{\pi}_1(i) b_{i,o_1}$ [Matlab Notation: $\underline{\alpha}_1 = \underline{\pi}_1 .* \underline{B}(:, o_1)'$]
- $t \rightarrow t+1$ ($t=1, 2, \dots, T-1$):

$$\begin{aligned} \underline{\alpha}_{t+1}(i) &= \left(\sum_{j \in E} \underline{\alpha}_t(j) p_{ji} \right) \Pr(O_{t+1}=o_{t+1} | X_{t+1}=i) \\ \underline{\alpha}_{t+1} &= (\underline{\alpha}_t \underline{P}) .* \underline{B}(:, o_{t+1})' \end{aligned}$$

- Finally: $\Pr(\underline{O}=\underline{o}) = \sum_{j \in E} \underline{\alpha}_T(j)$
- Computation can be illustrated in Trellis structure

□ Similarly (and identifiers needed later): Backwards procedure

- $\underline{\beta}_t := [\Pr(O_{t+1}=o_{t+1}, \dots, O_T=o_T | \mathbf{X}_t=1), \dots, \Pr(O_{t+1}=o_{t+1}, \dots, O_T=o_T | \mathbf{X}_t=N)]$
- $\underline{\beta}_T = \underline{1}$ (vector with all elements = 1); $\underline{\beta}_t = (\underline{P} * \underline{B}(:, o_{t+1}))' .* \underline{\beta}_{t+1}$

HMMs: Computations – most likely state sequence

Problem 2: Find 'most likely' state sequence for an observation $\underline{o}=[o_1, \dots, o_T]$ in a given HMM.

□ I.e. find the sequence $\underline{q}^*=[q_1^*, \dots, q_T^*]$ that maximizes $\Pr(X_1=q_1, \dots, X_T=q_T \mid \underline{O}=\underline{o})$ (or, equivalently, the joint probability)

□ Optimization via pre-fix of length t (Viterbi Algorithm):

$$\underline{\delta}_t := \left[\max_{q_1, \dots, q_{t-1}} \{ \Pr(X_1=q_1, \dots, X_{t-1}=q_{t-1}, \mathbf{X}_t=1, O_1=o_1, \dots, O_t=o_t) \}, \right. \\ \left. \dots, \max_{q_1, \dots, q_{t-1}} \{ \Pr(X_1=q_1, \dots, X_{t-1}=q_{t-1}, \mathbf{X}_t=N, O_1=o_1, \dots, O_t=o_t) \} \right]$$

□ Algorithm

■ $\underline{\delta}_1 = \underline{\pi}_1 \cdot \mathbf{B}(:, o_1)'$

■ $\underline{\delta}_{t+1}(j) = [\max_{i=1, \dots, N} \underline{\delta}_t(i) p_{ij}] B_{j, o_{t+1}}, \Psi_{t+1}(j) = \operatorname{argmax}_{i=1, \dots, N} \underline{\delta}_t(i) p_{ij}, \quad t=1, 2, \dots, T-1$

■ Maximum of probability: $p^* = \max_{i=1, \dots, N} \underline{\delta}_T(i), \quad q_T^* = \operatorname{argmax}_{i=1, \dots, N} \underline{\delta}_T(i)$

■ state sequence: $q_t^* = \Psi_{t+1}(q_{t+1}^*), \quad t=T-1, \dots, 1$

□ Efficient implementations: use of logarithms to avoid multiplications

HMMs: Computations –HMM fitting – not treated

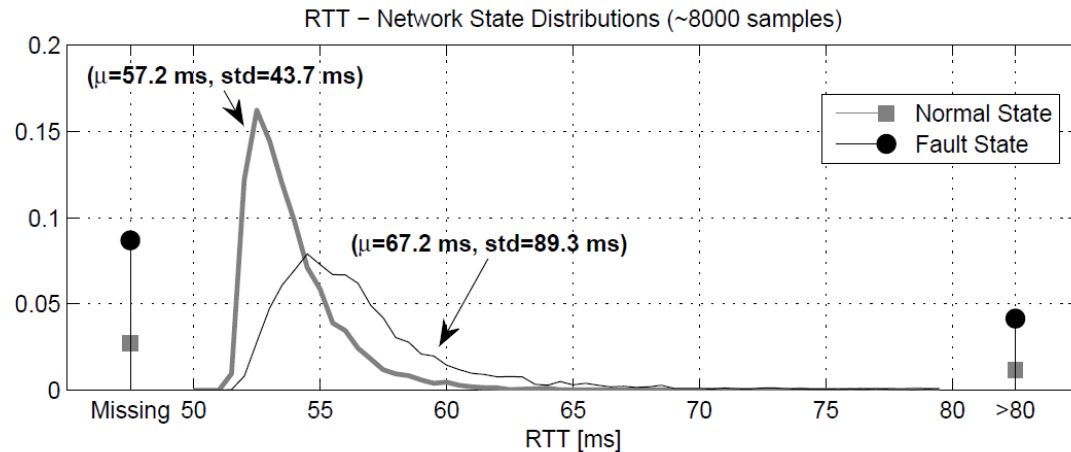
Problem 3: Find 'most likely' HMM model for an observation $\underline{o}=[o_1, \dots, o_T]$.

- Assumption: State-space E and symbol alphabet V are given
- Hence, desired is $\langle \underline{\pi}_1^*, \underline{P}^*, \underline{B}^* \rangle$ such that $\Pr_{\langle \underline{\pi}_1, \underline{P}, \underline{B} \rangle} (\underline{Q}=\underline{o})$ is maximized
- Iterative procedure for maximization: $\langle \underline{\pi}_1^{(m)}, \underline{P}^{(m)}, \underline{B}^{(m)} \rangle \rightarrow \langle \underline{\pi}_1^{(m+1)}, \underline{P}^{(m+1)}, \underline{B}^{(m+1)} \rangle$
 - Compute using model $\langle \underline{\pi}_1^{(m)}, \underline{P}^{(m)}, \underline{B}^{(m)} \rangle$:
 - $\gamma_t(i) := \Pr(X_t=i \mid \underline{Q}=\underline{o}) = \underline{\alpha}_t(i) \underline{\beta}_t(i) / \sum_i [\underline{\alpha}_t(i) \underline{\beta}_t(i)]$
 - $\xi_t(i,j) := \Pr(X_t=i, X_{t+1}=j \mid \underline{Q}=\underline{o}) = \underline{\alpha}_t(i) p_{ij} b_{j,o_{t+1}} \underline{\beta}_{t+1}(j) / \sum_j \sum_i [\underline{\alpha}_t(i) p_{ij} b_{j,o_{t+1}} \underline{\beta}_{t+1}(j)]$
 - 'Expectations':
 - $T(i) := \sum_{t=1}^{T-1} \gamma_t(i)$ = expected number of transitions from state i in $\underline{o}$
 - $T(i,j) := \sum_{t=1}^{T-1} \xi_t(i,j)$ = expected number of transitions from state i to state j in $\underline{o}$
 - $S(i,k) := \sum_{t=1, \dots, T, o_t=k} \gamma_t(i)$ = expected number of times in state i in $\underline{o}$ and observing symbol k
 - $S(i) := \sum_{t=1, \dots, T} \gamma_t(i)$ = expected number of times in state i in $\underline{o}$
 - Updated HMM:
 - $\underline{\pi}_1^{(m+1)} = [\gamma_1(1), \dots, \gamma_1(N)]$, $p_{ij}^{(m+1)} = T(i,j)/T(i)$,
 - $b_{ik}^{(m+1)} = S(i,k)/S(i)$
 - Update-step increases $\Pr_{\langle \underline{\pi}_1, \underline{P}, \underline{B} \rangle} (\underline{Q}=\underline{o})$, but potentially convergence to local maximum

HMM application: congestion detector

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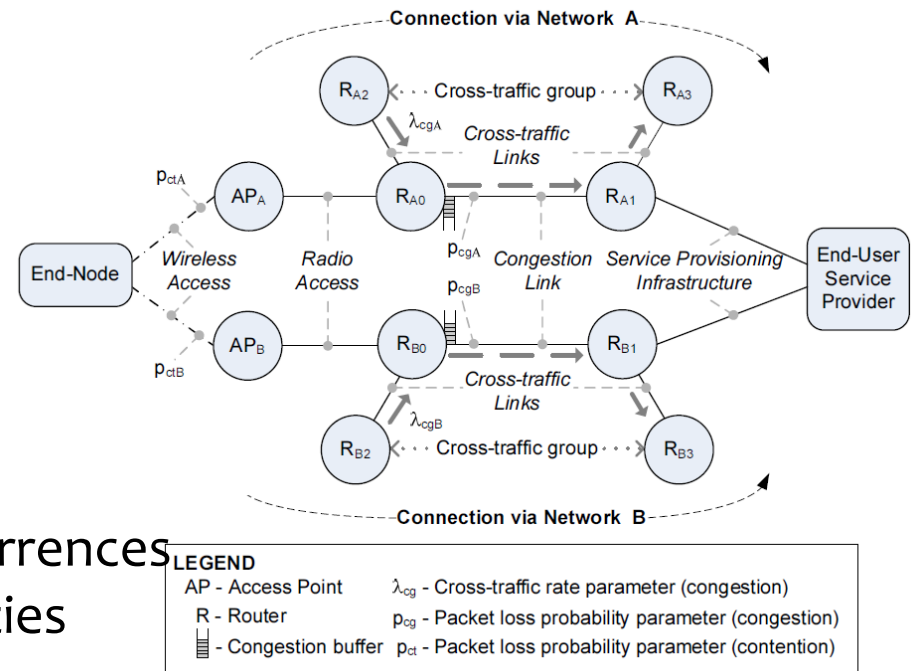
- Markov model for congestion levels
 - ▣ E.g., birth-death Markov process
 - ▣ Measurement of RTT distributions per congestion state
 - ▣ Discretized, e.g. to 5ms bins
- HMM model application
 - ▣ Discretize Observation sequence of RTTs
 - ▣ Calculate most likely state sequence (Viterbi algorithm)



HMM application: multiple faults and observations

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- Two fault-types:
 - ▣ Poor wireless link
 - ▣ Congestion in core network
- Modeled e.g. by 4-state Markov chain
 - ▣ Measurements on fault-occurrences for state transition probabilities
- 2 observations: RTT distribution and RSSI distribution measured for each state (each combination of faults)
- Discretized distributions mapped to HMM



Key points

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- What are the elements of a discrete time Markov Model?
 - ▣ states in model
 - ▣ transition probabilities

- What are the elements of an HMM?
 - ▣ Discrete time Markov Model (state evolutions)
 - ▣ Outputs/events/emissions → observation sequence O
 - output/emission probabilities

- What are the three main problems for HMMs?
 - ▣ 1) Compute probability of observation sequence.
 - ▣ 2) Determine most likely state sequence for observation sequence.
 - ▣ 3) Determine model parameters that maximizes probability of an observation

Learning outcomes

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- Be able to design an HMM-based fault-detector
 - ▣ Know the **elements** that comprise an HMM
 - ▣ Be able to **solve** simple discrete time HMM problems
 - Become familiar with **Forward** and **Viterbi** algorithms and how they work
 - ▣ Know how to map a fault-detection problem to an HMM-based algorithm

Exercises 2

Hidden Markov Models

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Given is the following 3-state hidden Markov model with parameters $\pi_1=[0.2,0.3,0.5]$, $P=[0.2,0.4,0.4; 0.5,0.1,0.4; 0.2,0.2,0.6]$. The observations are coin-toss results (Heads=1, Tails=2) with $B=[0.8,0.2;0.5,0.5;0.1,0.9]$. (This is an extended version of the numerical example from the lecture with three instead of two coins.)

- a) Write a (Matlab) program that generates observation sequences of length T from the given HMM.
- b) Write a program that efficiently computes the probability of a given observation sequence. Run the program for $S='HHTHTTTHT'$. Compare with a probability estimate via simulation using the program from question (a).
- c) Write a program to determine the most-likely state sequence and run the program for the sequence in question (b).
- d) (extra) Write a program to fit the HMM probabilities to the sequence. Run the program for S^5 (concatenating S with itself five times). Vary the starting probability vector and matrices and discuss the result.

Exercises - optional

Congestion detector

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- Assume a congestion model with 3 states, where the transitions form a birth-death Markov process with probabilities $p_{up}=0.2$, $p_{down}=0.2$; $p_{stay}=0.6$. For the first and last state, p_{stay} is increased by the probability of the not applied transition. A measurement of the RTTs in discretized intervals of 100ms results in the following probability values
 - ▣ State 1: [0.2,0.6,0.1, 0.05, 0.05, 0,0]
 - ▣ State 2: [0.1,0.2,0.3, 0.2, 0.1, 0.05, 0.05]
 - ▣ State 3: [0, 0.1, 0.1, 0.1, 0.3, 0.3, 0.1] ; State 3 is considered to be the fault state!
- Simulate RTT samples from this Hidden Markov process. Apply a threshold estimator to the sample sequence and plot the ROC curve.
- Calculate the point in the ROC space that is represented by the HMM-based estimator for this RTT sample sequence.

References

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- L. Rabiner, A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition, Proc. of IEEE, vol. 77, no. 2, February 1989.
- Philip Jackson's HMM tutorials 1 and 2 from:
<http://personal.ee.surrey.ac.uk/Personal/P.Jackson/tutorial/>